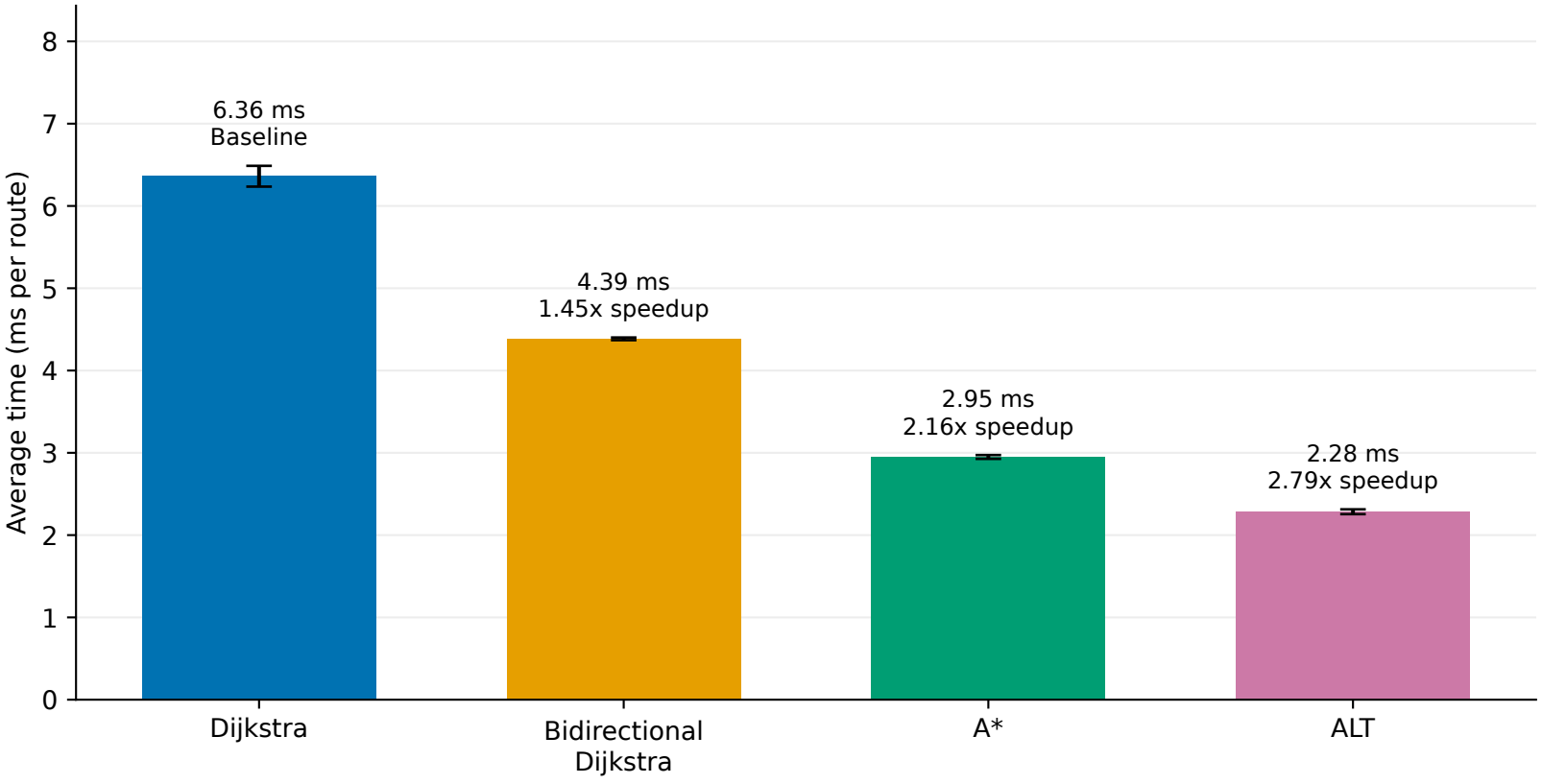

NY: 100 fixed routes (33 short, 33 medium, 34 long); ALT: 4 random landmarks, seed 2026.
Error bars: JMH 99.9% confidence intervals (3 forks, 10 measurement iterations per fork). Ratios compare means.